



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	WEB TECHNOLOGY AND APPLICATION
COURSE CODE	:	ICT600
EXAMINATION	:	JULY 2025
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts :
PART A (5 Questions)
PART B (2 Questions)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A**QUESTION 1**

a) Define the following web technology components.

- The Internet
- Web Standards
- Web Server
- Content Management System.

(8 marks)

b) Identify **FIVE (5)** key differences between a Web Browser and a Web Application.

(10 marks)

QUESTION 2

a) Define the following terms:

- HTML
- CSS
- Web Development

(6 marks)

b) State **THREE (3)** key principles of good web design.

(6 marks)

QUESTION 3

a) You are given a relational database table named Students with the following structure:

Students (StudentID INT, Name VARCHAR(100), Age INT, Major VARCHAR(100))

Prepare **FOUR (4)** SQL statements that demonstrate the CRUD operations (Create, Read, Update, Delete) on the above table.

(8 marks)

b) Produce a SQL query to display the names and majors of students who are enrolled in any course during the semester '2025S1', based on the following tables:

- Students (StudentID, Name, Major)
- Enrollments (EnrollmentID, StudentID, CourseCode, Semester)

(10 marks)

QUESTION 4

- a) You are developing a web application using a web framework such as Django or Laravel. Interpret how this framework helps to simplify web development compared to building the application from scratch using only raw HTML, CSS, JavaScript, and server-side scripting.
(4 marks)
- b) Using the Model-View-Controller (MVC) pattern, discover how a request to display a list of books would be handled in a web application. Apply an example for each Model, View, and Controller components would do in this scenario.
(6 marks)

QUESTION 5

- a) You are assigned to review a web application that accepts user input in search forms. Discover **THREE (3)** potential security risks using security and risk principles and suggest how each can be mitigated.
(6 marks)
- b) You are developing a web application that handles sensitive user data. Apply security coding practices to describe **THREE (3)** techniques you would implement to prevent common web vulnerabilities such as SQL injection, cross-site scripting (XSS), and broken authentication.
(6 marks)

PART B**QUESTION 1**

You are given the following HTML form:

```
<form id="registerForm">
<input type="text" id="username" placeholder="Enter Username" />
<input type="email" id="email" placeholder="Enter Email" />
<button type="submit">Register</button>
</form>
<div id="message"></div>
```

Construct JavaScript or jQuery code to validate the form such that:

- Both fields (username and email) must not be empty
- The email must follow a valid email format
- If valid, show message: "Registration Successful"
- If invalid, display a relevant error message in #message

Critically explain your chosen approach (JavaScript or jQuery) by comparing it with the alternative. Your explanation should include:

- Syntax and readability
- Maintainability and scalability
- Suitability for beginners
- Final recommendation on which to use in larger projects and why.

(15 marks)

QUESTION 2

Recommend a simple web-based program using a scripting language (e.g., JavaScript or PHP) for a Student Result Analyzer. The system should:

- Accept the student's name and an array of 5 numerical marks (e.g., [75, 64, 80, 55, 90])
- Calculate the total and average marks
- Determine and display a grade based on the average:
A (≥ 80), B (70–79), C (60–69), D (50–59), F (< 50)
- Display the student's name, all marks, total, average, and grade.

You must use appropriate variables, operators, strings, arrays, control structures, and functions in your solution.

(15 marks)

END OF QUESTION PAPER

